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Oral device to inhibit mastication and ingestion of solid food

Abstract

The present invention is oral device that inhibits mastication and ingestion of solid food while minimizing intrusiveness and discomfort in other oral activities. This device is ideal for individuals following an all-liquid diet without relying on willpower alone. The oral device allows a full range of motion of the jaw and the wearer keeps the ability to open and close the mouth, allowing all oral functions other than chewing and swallowing solid food. In various embodiments of the invention, the oral device comprises a set of retainers worn over teeth or retainer bars bonded to, or integrated as part of the retainers. A set of flexible mastication barriers coupled to the retainers or retainer bars and extending from mandible to the maxilla on the lingual side of the mouth inhibit the movement of solid food to the occlusal surfaces of the posterior teeth for chewing. An artificial intelligence platform is implemented to improve fabrication of the oral device and predict and eliminate undercuts in the retainers or retainer bars.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of Invention

(1) The present invention relates to oral devices that inhibit a wearer's ability to masticate and ingest solid food while allowing natural jaw movement without discomfort.

2. Description of Related Art

(2) An all-liquid diet has proven to be an effective diet for weight loss. This diet may be based upon any appropriate liquid or combination of liquids, such that dieters consume all the essential nutrients for a healthy lifestyle without ingredients that contribute to weight gain or prevent weight loss. Many solids, including fast food, commercially processed food, and food with added sugar, have been linked to weight gain, obesity, and health problems. A challenging aspect of maintaining a liquid diet is the continuous self-control needed to prevent consuming these solid foods.

(3) Mastication, i.e., chewing food into smaller particles, is essential for ingesting solid foods. Outside of wiring one's mouth shut, which increases the risk of choking and dental problems, attempts have been made to block solid foods from being swallowed while allowing tongue and jaw movement for talking. For example, U.S. Pat. No. 4,471,771 teaches a sieve-like blocking means secured to a user's upper teeth and pivots between closed and open positions. In its closed position, liquids and finely ground foods may freely pass through, but solid foods may not. However, during regurgitation, matter expelled from the stomach forces the blocking means to an open position, allowing the regurgitated matter to exit the mouth. Such a moving oral device is awkward and uncomfortable to wear, permits food to be trapped in the mouth, and does not inhibit the chewing of solid foods.

(4) U.S. Pat. No. 4,738,259 discloses an outwardly projecting flange attached to a support collar to

make chewing more difficult. Yet, this device does not entirely block food from contacting the tooth surfaces responsible for chewing and grinding solid food. In addition, the device is cumbersome and uncomfortable to wear and can be removed by the wearer.

(5) United States Patent Application Publication No. 2003/0075186 provides a device that restricts the opening of the wearer's mouth to slow the ingestion of food and purportedly increase satiety. However, it does not prevent the chewing and swallowing of solid foods. Moreover, a wearer would likely find the device impractical.

(6) Prior art devices like those above attach to the user's teeth in a manner that can cause teeth to shift, potentially resulting in a super eruption of teeth from the dental bones and increasing the likelihood of tooth damage or loss.

(7) A need exists for an oral weight control device that affixes to the teeth in a manner that does not cause the teeth to shift and does not impede tongue and jaw movement, thereby allowing the wearer to speak and otherwise open and close their mouth without restriction, while still inhibiting the chewing and swallowing of solid foods. Further, there is a need for such a device that minimizes the effect on the user, including discomfort and the difficulty of maintaining proper dental hygiene with such a device in place.

(8) The present inventor's U.S. Pat. Nos. 11,116,652 and 11,759,344, the entire disclosures of which are incorporated herein by reference, teach a dental appliance that utilizes an upper retainer and a lower retainer with left and right flexible food barriers on the lingual side of the teeth to prevent food that a user may attempt to masticate from being passed to the tongue to ingest. The dental appliance is comfortable and allows a full range of jaw motion. The wearer can perform all the mouth functions besides chewing and swallowing solid food. The present invention builds upon and expands the scope of the inventor's technology in several novel and non-obvious ways, as described below.

SUMMARY OF THE INVENTION

(9) The present invention provides an oral device that inhibits chewing and ingesting solid food while minimizing intrusiveness and discomfort in other oral activities. This device is ideal for individuals following an all-liquid diet without relying on willpower alone. The oral device allows a full range of motion of the jaw, and the wearer keeps the ability to open and close the mouth, allowing all oral functions other than chewing and swallowing solid food. In various embodiments of the invention, the oral device comprises a left and right set of retainer bars bonded to, or integrated as part of a retainer worn over, posterior teeth (molars and premolars/canines) to inhibit the movement of solid food to the occlusal surfaces of the posterior teeth for chewing. Each set of retainer bars is coupled to a flexible mastication barrier, extending from the mandible to the maxilla on the lingual side. Artificial intelligence is implemented to improve the fabrication of the oral device and predict and eliminate undercuts.

(10) In an embodiment of the invention, an oral device comprises a first mandibular retainer bar comprising a teeth contact surface contoured to a lingual surface of one or more mandibular posterior teeth on a first side of a recipient's mouth, a first maxillary retainer bar comprising a teeth contact surface contoured to a lingual surface of one or more maxillary posterior teeth on the first side of the recipient's mouth, and a first mastication barrier configured to be connected to the first mandibular retainer bar and the first maxillary retainer bar. The first mastication barrier comprises a first tab at a first edge and a second tab at a second edge opposite the first edge. The first mandibular retainer bar comprises a slot configured to receive the first tab of the first mastication barrier. The first maxillary retainer bar comprises a slot configured to receive the second tab of the first mastication barrier. The first tab and the second tab may be hollow cylinders. The first mandibular retainer bar comprises a handle arm configured to receive the first tab of the first mastication barrier. The first maxillary retainer bar comprises a handle arm configured to receive the second tab of the first mastication barrier. The first mandibular retainer bar may comprise a first array of hooks, and the first maxillary retainer bar comprises a second array of hooks. The first

mastication barrier may comprise a first array of loops at a first edge and a second array of loops at a second edge opposite the first edge. The second mandibular retainer bar comprises a teeth contact surface contoured to a lingual surface of one or more mandibular posterior teeth on a second side of the recipient's mouth, wherein the first side of the recipient's mouth is a left side and the second side of the recipient's mouth is a right side, a second maxillary retainer bar comprises a teeth contact surface contoured to a lingual surface of one or more maxillary posterior teeth on the second side of the recipient's mouth. A second mastication barrier is configured to connect to the second mandibular and the second maxillary retainer bars. The oral device may further comprise a first removable jig comprising a jig check including a surface contoured to a portion of the lingual surface, an occlusal surface, and a portion of a buccal surface of the one or more mandibular posterior teeth on the first side of the recipient's mouth, and a retainer holder including an arm configured to place the first mandibular retainer in a predetermined position adjacent to the lingual surface of the one or more mandibular posterior teeth on the first side of a recipient's mouth when the jig check is fitted over the lingual surface, an occlusal surface, and a portion of a buccal surface of the one or more mandibular posterior teeth on the first side of the recipient's mouth.

(11) In another embodiment of the invention, a method for inhibiting mastication of solid food comprises the steps of bonding a first mandibular retainer bar to a lingual surface of one or more mandibular posterior teeth on a first side of a recipient's mouth, bonding a first maxillary retainer bar to a lingual surface of one or more maxillary posterior teeth on the first side of a recipient's mouth, and connecting a first mastication barrier to the first mandibular retainer bar and the second maxillary retainer bar. The step of connecting comprises inserting a first tab of the first mastication barrier into a slot of the first mandibular retainer bar and inserting a second tab of the first mastication barrier into a slot of the first maxillary retainer bar. The may further comprise the steps of bonding a second mandibular retainer bar to a lingual surface of one or more mandibular posterior teeth on a second side of the recipient's mouth, wherein the first side of the recipient's mouth is a left side and the second side of the recipient's mouth is a right side, bonding a second maxillary retainer bar to a lingual surface of one or more maxillary posterior teeth on the second side of the recipient's mouth, and connecting a second mastication barrier configured to the second mandibular retainer bar and the second maxillary retainer bar. The step of bonding the first mandibular retainer bar to the lingual surface of the one or more mandibular posterior teeth on the first side of the recipient's mouth comprises placing a jig check over a portion of the lingual surface, an occlusal surface, and a portion of a buccal surface of the one or more mandibular posterior teeth on the first side of a recipient's mouth.

(12) In another embodiment of the invention, an oral device comprises a jig check including a surface contoured to a portion of a lingual surface, an occlusal surface, and a portion of a buccal surface of one or more posterior teeth, and a retainer holder including an arm configured to place a retainer bar in a predetermined position adjacent to the lingual surface of the one or more posterior teeth when the jig check is placed over the portion of the lingual surface, the occlusal surface, and the portion of the buccal surface of the one or more posterior teeth.

(13) In yet another embodiment of the invention, an oral device comprises: a first retainer configured to fit over a recipient's mandibular teeth, wherein the first retainer comprises a first retainer bar adjacent to a lingual surface of one or more mandibular posterior teeth on a first side of the recipient's mouth, a second retainer bar adjacent to a lingual surface of one or more mandibular posterior teeth on a second side of the recipient's mouth, a second retainer configured to fit over the recipient's maxillary teeth, wherein the second retainer comprises a first retainer bar adjacent to a lingual surface of one or more maxillary posterior teeth on the first side of the recipient's mouth, a second retainer bar adjacent to a lingual surface of one or more maxillary posterior teeth on the second side of the recipient's mouth, a first mastication barrier configured to be connected to the first retainer bar of the first retainer and the first retainer bar of the second retainer, and a second mastication barrier configured to be connected to the second retainer bar of the first retainer and the

second retainer bar of the second retainer. The first mastication barrier comprises a first tab at a first edge and a second tab at a second edge opposite the first edge. The first mandibular retainer bar comprises a slot configured to receive the first tab of the first mastication barrier. The first maxillary retainer bar comprises a slot configured to receive the second tab of the first mastication barrier.

(14) In another embodiment of the invention, a method of creating a retainer comprises the steps of receiving a dental scan representing a dental structure; processing the dental scan using an artificial intelligence model, wherein the artificial intelligence model is trained on a dataset comprising a plurality of dental scans representing unique dental structures; and generating a digital model of a dental retainer based on the processed dental scan, wherein the digital model is optimized, through the artificial intelligence model, for the dental structure represented in the dental scan by eliminating one or more undercuts. The method may further comprise the step of fabricating the dental retainer based on the digital model using a manufacturing device. The artificial intelligence model comprises a machine learning model, a deep learning model, or a combination thereof. The unique dental structures comprise various dental structures, dental conditions, or a combination thereof. The manufacturing device may comprise a 3D printer. The method may further comprise the step of adjusting the digital model based on one or more corrections received from a user or a dental professional.

(15) The present invention eliminates material occupying the oral cavity and is aesthetically pleasing over the prior art. It also increases the accessibility of teeth for hygiene and minimizes discomfort. Due to its reduced size, the present invention does not interfere with the patient's speech or tongue space. The present invention is efficient to manufacture, and the resulting oral device is relatively easy to install in a wearer's mouth.

(16) Other devices, apparatus, systems, methods, features, and advantages of the invention will become apparent to one with skill in the art upon examination of the following figures and detailed description. It is intended that all such additional systems, methods, features, and advantages be included within this description, as well as the scope of the invention, and protected by the accompanying claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a complete understanding of the present invention and its advantages, reference is now made to the ensuing descriptions taken in connection with the accompanying drawings briefly described as follows.

(2) FIG. 1 illustrates a set of retainer bars for a left side of a mouth according to an embodiment of the invention.

(3) FIG. 2 illustrates a mandible with the mandibular retainer bar secured to mandibular posterior teeth according to an embodiment of the invention.

(4) FIG. 3 illustrates a complete set of retainer bars according to an embodiment of the invention.

(5) FIG. 4 illustrates a close-up view of the set of retainer bars for the left side of the mouth.

(6) FIG. 5 illustrates a mastication barrier according to an embodiment of the invention.

(7) FIG. 6 is an end view of the flexible barrier and its dimensions according to an exemplary embodiment of the invention.

(8) FIG. 7 is a top view of the flexible barrier and its dimensions according to an exemplary embodiment of the invention.

(9) FIG. 8 illustrates a rear view of a complete oral device according to an embodiment of the invention in a replica of a user's mandible and maxilla.

(10) FIG. 9 illustrates a side-rear view of the complete oral device shown in FIG. 8.

- (11) FIG. 10 illustrates a mastication barrier 500 and a retainer bar 110 in a disassembled state.
- (12) FIG. 11 illustrates a tab of the mastication barrier partially inserted into the slot of the retainer bar of FIG. 10.
- (13) FIG. 12 illustrates the mastication barrier fully inserted into the retainer bar of FIG. 10.
- (14) FIG. 13 illustrates an optional jig according to an embodiment of the invention.
- (15) FIG. 14 depicts the optional jig of FIG. 13 as transparent to show the teeth and retainer bar underneath.
- (16) FIG. 15 illustrates an aligner for anterior teeth according to an embodiment of the invention.
- (17) FIG. 16 illustrates a retainer bar according to another embodiment of the invention.
- (18) FIG. 17 illustrates a mastication barrier for use with the retainer bar of FIG. 16.
- (19) FIG. 18 illustrates a top view of a retainer bar according to another embodiment of the invention.
- (20) FIG. 19 illustrates a bottom view of the retainer bar of FIG. 18.
- (21) FIG. 20 illustrates a segmented retainer bar according to another embodiment of the invention.
- (22) FIG. 21 illustrates a retainer system according to an embodiment of the invention.
- (23) FIG. 22 illustrates an oral device according to an embodiment of the invention.
- (24) FIG. 23 illustrates a retainer system according to an embodiment of the invention.
- (25) FIG. 24 illustrates a two-component retainer bar connection system according to an embodiment of the invention.
- (26) FIG. 25 illustrates a retainer attachment system according to an embodiment of the invention.
- (27) FIG. 26 illustrates a retainer system according to an embodiment of the invention.
- (28) FIG. 27 illustrates a schematic of an artificial intelligence platform according to an embodiment of the invention.
- (29) FIG. 28 illustrates an initial pass at filling in undercuts.
- (30) FIG. 29 illustrates a second pass at filling in undercuts.
- (31) FIG. 30 illustrates a final pass at filling in undercuts.
- (32) FIG. 31 illustrates a side view of delineating a full retainer's gum-adjacent outline.
- (33) FIG. 32 illustrates a front-to-back view of a full retainer's gum-adjacent outline delineation.
- (34) FIG. 33 illustrates a completed retainer design.
- (35) FIG. 34 illustrates a rendering of an exemplary retainer design in STL format ready for a 3D printer.
- (36) FIG. 35 illustrates a top-down view of hooks added to an exemplary retainer design.
- (37) FIG. 36 illustrates a side view of hooks added to an exemplary retainer design.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(38) Preferred embodiments of the present invention and their advantages may be understood by referring to FIGS. 1-36. Although the present invention is described in the context of inhibiting chewing and ingestion of solid food, the present invention may be used for any oral or orthodontic application where it is desirable to limit the movement of solid food with the mouth or restrict the movement of teeth.

(39) In general, the present invention is an oral device attached to a wearer's jaw to interfere with the positioning of food on the teeth, thereby frustrating the wearer's ability to chew and ingest solid foods. The oral device comprises a flexible or foldable physical barrier on each side of the mouth that prevents the tongue from passing solid foods from the lingual portion of the oral cavity to the occlusal portion. These mastication barriers prevent food from moving between the occlusal surfaces of teeth, usually masticated by the premolars and molars, from moving back into the oral cavity, where it could be swallowed. Form-fitted retainers attached to or fitted over mandibular and maxillary teeth position the flexible food barriers on each lingual side of the oral cavity.

(40) The embodiments of the present invention are advantageous in maintaining an all-liquid diet without impeding a wearer's ability to open and close the mouth for activities other than chewing, such as speaking, breathing, and yawning. There is no need to fully cover the teeth or utilize

material on the buccal side of teeth, eliminating irritation to the user's lip and cheek lining and allowing better hygiene.

(41) In one embodiment of the invention, the oral device comprises four retainer bars to position two flexible food barriers within the oral cavity. Two retainer bars are bonded to the mandible, and two more are bonded to the maxilla. Bonding is performed using dental cement or any other type of suitable adhesive, the identification and implementation of which are apparent to one of ordinary skill in the art, such as but not limited to dental composite resins. The term bonding cement, as used herein, includes not only dental cement but all types of suitable adhesives. Bonding is preferably semi-permanent, so the retainer bars may be removed later if desired. Each jaw set of retainers includes one retainer bar secured to the lingual surface of several teeth on the left side and one secured to the lingual surface of several teeth on the right side. In an embodiment of the invention, the teeth utilized for securing a respective retainer bar are limited to premolars and molars. The exact number of teeth used may vary depending on the number and size of teeth present and other oral parameters so long as the flexible barriers attached to it sufficiently inhibit mastication on the occlusal surfaces of the posterior teeth. In other embodiments of the invention, anterior teeth may also be included to secure a retainer bar.

(42) As used herein, the term “bar” departs from its strict, plain, and ordinary meaning. The retainer bars of the present invention are not necessarily shaped like a long rod. They may be shaped in any manner to accommodate attachment to teeth and attachment to a corresponding flexible mastication barrier.

(43) The term “retainer,” as used herein, also includes a “tray” within its scope. In strict terminology, a retainer is a device worn to maintain a new position of teeth after an individual has undergone orthodontic treatment. A tray such as an Invisalign tray actively moves teeth and is not technically a retainer, though it may look similar. In the present invention, a retainer may retain or actively move teeth, and thus, the scope of such includes a tray.

(44) Various components of the present invention may be manufactured from FDA-approved materials, the identification of which is apparent to one of ordinary skill in the art.

(45) FIG. 1 illustrates a set of retainer bars **100** for a left side of a mouth according to an embodiment of the invention. The set of left retainer bars **100** comprises a mandibular retainer bar **110** and a maxillary retainer bar **120**. The mandibular retainer bar **110** comprises a slot **112** and a slot entry **113**. The slot **112** and the slot entry **113** receive a tab of a flexible mastication barrier, the attachment of which is described below. In an optional embodiment of the invention, there may be one or more openings **114** to minimize material or assist in securing the respective retainer. The mandibular retainer bar **110** includes teeth contact surface **116** contoured to match the lingual surface of a particular user's corresponding posterior teeth. Accordingly, the teeth contact surface **116** is prepared from a dental scan of the recipient's mouth, the identification and implementation of which are apparent to one of ordinary skill in the art. For example, any intraoral scanner may be used, including but not limited to 3D video, confocal imaging, structured light, laser, and LED pattern projection with appropriate digital processing, and software may be used to generate a digital representation of the teeth and oral cavity. The digital output of the dental scan is used to manufacture a custom-fitted retainer bar contoured to the recipient's teeth in this and other embodiments described herein. Accordingly, variations in the recipient's teeth' shape, size, and layout are accounted for to ensure maximum surface contact between a retainer bar and the corresponding teeth. The maxillary retainer bar **120** also includes a slot **122**, a slot entry **123**, and openings **124**; however, these are hidden from view in this figure. They are shown in the following figures.

(46) FIG. 2 illustrates a mandible with the mandibular retainer bar **110** secured to mandibular posterior teeth according to an embodiment of the invention. Here, the contour matching of the teeth contact surface **116** to the lingual surface of the mandibular posterior teeth is shown. Contour matching via a dental scan permits the lingual gaps between teeth and between teeth and a

respective retainer bar to be occupied with retainer bar material. This gap-filling prevents the movement and ingestion of chewed food from the occlusal surfaces of the posterior teeth. Gap filling also better secures the respective retainer bars. Slot **112** is located on the top surface of the mandibular retainer bar **110**. Moreover, the corners of the mandibular retainer bar **110** are rounded, as shown to minimize discomfort associated with the movement of a user's tongue. Any rough edges or transitions between surfaces may also be smoothed.

(47) The right side of FIG. 2 depicts a set of retainer bars **200** for the right side of the mouth. The set of right retainer bars **200** comprises a mandibular retainer bar **210** and a maxillary retainer bar **220**. The set of right retainer bars **200** includes the same elements described and shown concerning the set of left retainer bars **100**, except that they are contoured and secured to the posterior teeth on the right side of the mouth. In FIG. 2, the mandibular retainer bar **210** is secured to the mandible. The mandibular retainer bar **210** includes a slot **212**, a slot entry **213**, an opening **214**, and a teeth contact surface **216**, shown in the following figures. To show its relative location when used with a corresponding flexible barrier, the maxillary retainer bar **220** is shown floating above the mandibular retainer bar, as the upper jaw is not included in this figure. The maxillary retainer bar **220** includes slot **222**, slot entry **223**, opening **224**, and teeth contact surface **226**, shown in the following figures.

(48) FIG. 3 illustrates a complete set of retainer bars **300** according to an embodiment of the invention. The complete set of retainer bars **300** comprises the set of retainer bars **100** for the left side of the mouth and the set of retainer bars **200** for the right side of the mouth. Here, slot **212** and slot entry **213** of the mandibular retainer bar **210** are better shown along with slot entry **223** for the maxillary retainer bar **220**. Although slot **212** is shown as positioned on the lingual and palatal side, slot position may vary on any of the retainer bars so long as it does not interfere with the attachment of the respective mastication barrier.

(49) FIG. 4 illustrates a close-up view of the set of retainer bars **100** for the left side of the mouth. The mandibular retainer bar **210** is secured to the mandibular posterior teeth. To show its relative location when used with a corresponding flexible barrier, the maxillary retainer bar **220** is shown floating above the mandibular retainer bar **210**, as the upper jaw is not included in this figure.

(50) FIG. 5 illustrates a mastication barrier **500** according to an embodiment of the invention. A mastication barrier **500** will be used on each side of the mouth, i.e., in both the set of retainer bars **100** for the left side and the set of retainer bars **200** for the right side. The dimensions of the mastication barrier **500** may vary between the two sets to accommodate any mouth asymmetry. The mastication barrier **500** comprises a body **510** disposed between a tab **512** at one edge and another at an opposite edge. Each tab **512** slides into a respective slot **112**, **122**, **212**, or **222** of a set of retainer bars **100** or **200** via corresponding openings **113**, **123**, **213**, or **223**. In other words, slots **112**, **122**, **212**, and **222** and tabs **512** correspond to female and male connectors. This slot and tab configuration is easy for a dental professional to install and is comfortable for the wearer.

(51) The mastication barrier **500** is preferably a deformable elastic material to form a pressure-fit connection with the retainer bars. In an embodiment of the invention, the body **510** and tabs **512** are integrated as a single material. Exemplary materials include but are not limited to neoprene, silicone, and synthetic or natural rubber. However, more rigid materials such as polymers, ceramics, metals, and metal alloys may be used, including materials often used in dental implants like zirconia. These materials may also be used in the other embodiments described below. More preferably, the body **510** experiences minimal or no folding when the user's mouth is closed, which allows the user to open their mouth without any undue effort. In other embodiments, the body and tabs **512** are two materials joined together. For example, tabs **512** may be formed from a sufficiently rigid or non-deforming material, facilitating their seamless insertion into the slots of the retainer bars.

(52) FIGS. 6 and 7 illustrate the dimensions of the flexible barrier **500** according to an exemplary embodiment of the invention. For example, the tabs **512** are cylindrical with a diameter of 1.95-

2.00 mm. The slots will have a corresponding but slightly smaller diameter (e.g., a connector diameter of 1.998 mm for a cylindrical cavity diameter of 2.000 mm). Different geometries may be implemented for retainer bar attachment in other embodiments, such as square, rectangular, hexagon, or triangular shapes. Such geometries may be slid into place in matching counterpart slots or have an alternative locking geometry that a dental professional can secure during placement.

(53) FIGS. 8 and 9 illustrate a complete oral device **800** according to an embodiment of the invention in a mockup jaw. The oral device **800** comprises a set of retainer bars **100** for the left side of the mouth and a set of retainer bars **200** for the right side of the mouth. The set of retainer bars **100** are coupled to one another through the flexible mastication barrier **500**. Likewise, the set of retainer bars **200** are coupled to one another through another flexible mastication barrier **500**. To enhance contrast, the barriers **500** are annotated in black. In practice, the retaining bars are designed to mimic the color of teeth, and transparent flexible barriers are used. The oral device **800** forms a physical barrier on each side of the mouth sufficient to prevent solid foods from reaching the occlusal surfaces of the posterior teeth, thereby compelling users to follow a liquid diet as solid food cannot be chewed and ingested.

(54) The length of each retaining bar can vary from a single tooth, approximately 7 mm, to four teeth, approximately 30 mm, depending on the dimensions of the wearer's posterior teeth. In one exemplary embodiment, the retainer bars have external dimensions of 3-5 mm in height by 3.5-5 mm in width (extending mesially to distally in the lingual and palatal portions of the surface of the teeth to which the retainers/bars are attached), and generally 25-26 mm length for most users. The exact dimensions may be calibrated based on a scan of a user's oral anatomy such that there is a minimum 4.360 mm vertical clearance between the palatal and lingual retainers on each side and approximately 23.52 mm horizontal clearance between the left and right retainers inside the oral cavity. This helps ensure adequate space for the wearer's tongue to rest and move for purposes other than chewing without discomfort.

(55) FIGS. 10-12 illustrate a mastication barrier **500** joining with a retainer bar **110**. The mastication barrier **500** is a rigid mockup and would otherwise have a flexible body **510** in practice. Referring to FIG. 10, the mastication barrier **500** and the retainer bar **110** are disassembled. Referring to FIG. 11, tab **512** of the mastication barrier **500** is inserted into slot **112** of the retainer bar **110** starting at opening **113**. Referring to FIG. 12, the mastication barrier **500** and the retainer bar **110** are fully assembled. Once in position, a dental professional may semi-permanently secure the tab **512** to retainer bar **110** by placing cement at holes **114**.

(56) FIG. 13 illustrates an optional jig **1300** according to an embodiment of the invention. The jig **1300** facilitates the accurate placement and attachment of the retaining bars to the teeth. The jig **1300** comprises two interlocking and removable pieces: a jig check **1310** and a retainer holder **1320**. The jig check **1310** fits onto the occlusal surface of the teeth and features a rectangular slot **1312**. The retainer holder **1320** is a U-shaped piece with two arms, **1322** and **1324**. The lower arm **1324** of the retainer holder **1320** secures a retainer bar, in this case, retainer bar **110** to the user's teeth, while the upper arm **1322** fits through the rectangular slot **1312** in the jig check **1310**. The bottom surface (or top surface if used on the maxilla) of the jig check **1310** is contoured to match the occlusal surface, a portion of the lingual surface, and a portion of the buccal surface of a portion (as shown) or all (not shown) of the posterior teeth for which the retainer bar **110** is to be attached. The contoured surface is prepared from a dental scan of the recipient's teeth. The dimensions of the jig check **1310**, upper arm **1322**, and lower arm **1324** are determined based on the dental scan and the desired placement of the retainer bar **110**.

(57) When fitted over the user's teeth, the jig check **1310** restricts movement of the posterior teeth underneath it and ensures proper vertical and lateral positioning of the respective retainer bar **110**. During placement, the jig check **1310** temporarily covers portions of both the lingual and buccal sides of the wearer's teeth, effectively locking the retainer bar **110** via the retainer holder **1320** in place for bonding by the dental professional. The jig **1300** is removed after the bonding cement is

adequately cured. To show the positioning of the jig **1300** over the posterior teeth, FIG. **14** depicts the optional jig **1300** as a transparent replica. A custom jig **1300** is prepared for each respective retainer bar **110**, **120**, **210**, and **220**. In an embodiment of the invention, the jig check and retainer holder are integrated as a single component.

(58) FIG. **15** illustrates an aligner **1500** according to an embodiment of the invention. Aligner **1500** includes a teeth surface **1510** contoured to the anterior teeth or anterior teeth and posterior teeth that are not bonded to the retainer bars utilized on the mandible or maxilla. The aligner **1500** is prepared from a dental scan of the teeth it covers and, when used, restrains or forces slight movement of the covered teeth in a desired position. As shown, the teeth surface **1510** is contoured to the respective teeth' lingual, occlusal, and buccal surfaces. Aligner **1500** fills the space of teeth between the retainer bars bonded to the mandible or maxilla. The aligner **1500** is removable and thus can be used as a temporary retainer during, for example, sleeping. Alternatively, the aligner can be worn at all times except during drinking, e.g., consuming a liquid diet. The aligner **1500** may be manufactured from a clear, rigid plastic.

(59) FIG. **16** illustrates a retainer bar **1600** according to another embodiment of the invention. The retainer bar **1600** comprises a handle arm **1610** and a teeth band **1620**. The handle arm **1610** is integrated into the teeth band **1620** through elbow **1612**. The handle arm **1610** extends from the elbow **1612** in a manner that is generally parallel to the teeth band **1620**. However, an opening **1614** is present between the handle arm **1610** and the teeth band **1620** on the handle arm **1610** end opposite the elbow **1612**. The teeth band **1620** includes sections **1611**, **1612**, and **1613** contoured to a portion of the lingual surface of the respective teeth, in this case, three posterior teeth; however, the actual number may vary in practice. The teeth sections **1611**, **1612**, and **1613** form a teeth contact surface developed from a dental scan of the respective teeth. Optionally, one or more of the tooth sections **1611** may serve as an arch between tooth sections. For example, teeth section **1612** may serve as an arch between tooth section **1611** and tooth section **1613**. The arch does not come in contact with the lingual surface of a tooth. For simplification, only one retainer bar **1600** is shown; however, four retainer bars will be used in the complete oral device: one for the mandibular posterior teeth on the left side of the mouth, one for the maxillary posterior teeth on the left side of the mouth, one for the mandibular posterior teeth on the right side of the mouth, and one for the maxillary posterior teeth on the right side of the mouth.

(60) FIG. **17** illustrates a mastication barrier **1700** for use with two retainer bars **1600**. Like mastication barrier **500**, the mastication barrier **1700** comprises a body **1710** and tabs **1720**. However, in this embodiment, the tabs **1720** are hollow cylinders configured to receive a handle arm **1610** snugly to secure the mastication barrier **1700** to two retainer bars **1600** on the left or right side of the recipient's mouth. The distal end of the handle arm **1610** may include a protrusion or notch at opening **1614** to prevent a tab **1720** from sliding off the handle arm **1610**.

(61) FIGS. **18** and **19** illustrate a retainer bar **1800** according to another embodiment of the invention. FIG. **18** illustrates a top view of the retainer bar **1800**, and FIG. **19** illustrates a bottom view. Here, the retainer bar **1800** comprises teeth contact surface **1810** on one side and an array of hooks **1820** on the opposite side. Here, the retainer bar **1800** is formed as an integrated single piece of material. The array of hooks **1820** comprises a number (N) of individual hooks **1821A-N** running along the length of the retainer bar **1800** to provide a fastening means for securing an edge of a mastication barrier, which is further described below. Hooks **1821A-N** may be L- or J-shaped. The number N of individual hooks **1821N** may be any number suitable for adequately securing a respective mastication barrier. The teeth contact surface **1810** is contoured using a dental scan to match the lingual surface of the respective posterior teeth for which it is secured. For simplification, only one retainer bar **1800** is shown, however four retainer bars will be used in the complete oral device as described above.

(62) FIG. **20** illustrates a segmented retainer bar **2000** according to another embodiment of the invention. The segmented retainer bar **2000** is similar to retainer bar **1800** in that it includes hooks;

however, it has been segmented into sections **2010**, **2020**, and **2030**. Each section **2010**, **2020**, or **2030** includes a tooth contact surface **2011**, **2021**, or **2031** and an array of hooks **2012**, **2022**, or **2032**. Each array of hooks comprises a number of individual hooks spaced along the section, sufficient to secure a portion of the edge of a mastication barrier. The advantage of the segmented retainer bar **2000** is that each section **2010**, **2020**, and **2030** may be cut from a length of continuous and flexible retainer bar material. The length of each section corresponds to a portion of the respective tooth's lingual circumference. Each flexible section is then bonded to the respective tooth, eliminating the need for a dental scan.

(63) FIG. **21** illustrates a retainer system **2100** according to another embodiment of the invention. The retainer system **2100** comprises a retainer **2105** with integrated retainer bars **2110** and **2120** on the respective lingual surface of each side of the mouth. Only one retainer **2105** is shown for the mandible. A complementary retainer (not shown) would be used for the maxilla.

(64) FIG. **22** illustrates an oral device **2200** according to an embodiment of the invention. Here, the oral device **220** comprises a pair of retainer systems **2100** connected through mastication barriers **2210**. The integrated retainer bars **2110** and **2120** are obstructed from view. The mastication barriers **2210** are flexible and stretch, allowing the user's mouth to open to the maximum biological extent. In one embodiment, minimal, if any, folding of the mastication barriers **2210** occurs when the mouth is closed. The degree of elasticity of each flexible mastication barrier **2210** may vary to provide different degrees of resistance. In another embodiment, the mastication barriers **2210** may fold toward the tongue or cheek, or folding may be limited between the teeth.

(65) FIG. **23** illustrates a retainer system **2300** according to an embodiment of the invention. Here, the retainer system **2300** comprises a retainer bar **2310** attached to the posterior teeth on the left side, a retainer bar **2320** attached to the posterior teeth on the right side, and a retainer bar **2330** attached to the anterior teeth of the mandible. A complementary retainer system (not shown) would be used for the maxilla. The retainer bars **2310** and **2320** are secured to a set of mastication barriers as described above. The retainer bar **2330** minimizes, if not wholly prevents, super-eruption of the teeth not secured to retainer bars **2310** and **2320**. In other words, the teeth not utilized for securing the mastication barriers are held in place by a separate retainer in this embodiment. In other embodiments, the retainer bar **2330** is used with a portion or all of the teeth not used for the retainer bars **2310** and **2320** and may include one or more posterior teeth or less than all of the anterior teeth.

(66) The retainer bar **2330** may be used as a standalone device. In one embodiment, it can be made of a thermoplastic material such as polyurethane, polyethylene terephthalate glycol (PETG), or polyvinyl chloride (PVC), which are known in commercial use for their durability, transparency, and biocompatibility. The anterior retainers are custom-made in one embodiment to fit the user's teeth. They are designed to be inconspicuous while providing the necessary support to prevent teeth from shifting back to their original positions.

(67) FIG. **24** illustrates a two-component retainer bar connection system **2400** according to an embodiment of the invention. Here, the retainer bar system **2400** comprises a first component **2410** with a teeth contact surface **2412** contoured to the user's teeth as described above and a second component **2420** that interlocks with the first component **2410**. On its opposite side, the first component **2410** has a set of male connectors **2414**. The second component **2420** includes a set of corresponding female connectors **2424**. When coupled, the male connectors **2414** and female connectors **2424** provide an interference fit to secure the edge of a mastication barrier between the first component **2410** and the second component **2420**. Such a connection system may also connect the opposite edge of the mastication barrier.

(68) FIG. **25** illustrates a retainer attachment system **2500** for attaching a retainer or retainer bar as described above according to an embodiment of the invention. The attachment system **2500** comprises one or more projections **2502** affixed to the user's teeth. A retainer **2510** (e.g., a full retainer as in FIG. **21** or, alternatively, one of the retainer bars described in the systems above)

comprises one or more correspondingly shaped notches **2512**. The projection **2502** and notch **2510** are configured to provide an interference fit to keep the retainer **2510** in place when pressed against the projection **2502**. Although the projection **2502** and corresponding notch **2512** are depicted as triangular, other geometric shapes may be used. In this embodiment, the base side of the triangular-shaped projection **2502** and notch **2512** is disposed near the occlusal surface of the teeth.

(69) FIG. **26** illustrates a retainer system **2600** according to an embodiment of the invention. Here, the retainer bar system **2600** comprises a retainer bar **2610** constructed of bands **2612** fitted around posterior teeth. Affixed to one or more bands **2612** is a bar **2614** to attach a mastication barrier. The bands **2612** may be manufactured from a solid material such as but not limited to metal or flexible material that can be friction-fitted over teeth. The bar **2614** is preferably rigid to provide a secure mount for a mastication barrier. Two or four retainer systems **2600** would be used to form a complete oral device.

(70) To improve fabrication and accuracy of the fit of retainers like retainer **2105** over the mandible or maxilla, an artificial intelligence (AI) platform based on a deep convolutional neural network (CNN or DCNN) architecture is implemented to predict and remove undercuts. An undercut is the cross-sectional portion of the retainer that prevents it from being inserted or fitted over the corresponding teeth. For example, many teeth are wider at their occlusal surface than at the gum. Before this invention, fabrication required a lengthy process of scanning teeth, sending the scans off-site, and then hand-creating custom-fit retainers. This process is also labor intensive and requires skilled dental lab technicians trained in design software such as Solidworks or 3Shape and 3D printing. To streamline this process most accurately, the present invention rapidly corrects and fills in undercuts around the teeth and gums and between the teeth in the interproximal areas and black triangles to avoid locking in the retainers/appliances. It also corrects any distortions in the scans and thus designs the retainers and attachments and outputs data files to facilitate 3D-printed with precision to ensure an optimal fit and user comfort. AI printing of retainer pieces and barriers has yet to be implemented in commercial practice.

(71) Utilizing an AI platform for design can ensure improved fit in retainers over traditional laboratory techniques, which facilitates attachment of retainer pieces to a user's teeth with less effort and finesse required on the part of the dental professional. Further, using AI for design reduces fabrication costs and savings, which can be passed to consumers to make the invention, facilitating a weight loss program available to all economic classes.

(72) Though sophisticated software is used, every dimension and feature must be precisely defined using complex, domain-specific software tools to produce a usable design ready for manufacturing. FIG. **27** depicts a schematic of a DCNN architecture according to an embodiment of the invention. AI algorithms are created and trained to design the retainer bars and barriers to implement this structure. The designed retainers and barriers are produced in STL format and then sent to a 3D printer and printed. This forms a custom-fit retainer for each user.

(73) In an embodiment of the invention, the AI algorithms are trained using 500-1,000 scans of the target dentition, covering both the upper and lower arches prepared with the undercuts present in natural dentition being strategically filled in. Teeth are anatomically unique, each serving specific functions in chewing, speaking, and aesthetics, which makes it challenging to create a generic retainer that fits all. Currently, each retainer is individually designed by a lab technician, manually or using computer programs, which is a labor-intensive process. This AI platform aims to streamline the retainer design process using a large dataset to train the platform through machine learning. This will simplify the design process by addressing issues such as undercuts-areas where teeth widen on the chewing surface and narrow towards the gumline, potentially causing plastics to get stuck. Lab technicians currently spend considerable time smoothing these undercuts to facilitate easier placement and removal of retainers. The platform aims to automate and optimize the retainer design process by utilizing machine learning, improving efficiency and precision in creating custom-fit retainers.

(74) As the algorithms used in AI learning improve, the automated identification of the undercuts becomes faster and more accurate. After the 500-1,000 training scans are completed, the AI can efficiently eliminate undercuts and establish appropriate borders for retainer fabrication. Using AI to complete this process allows the time of processing to be reduced from a matter of hours to a matter of seconds. The progression of this learning can be observed in FIG. 28, which demonstrates a first pass by the AI at filling in the undercuts; FIG. 29, a second pass; and FIG. 30, a final pass. Once the processing of FIGS. 28-30 was completed, and the delineation of the retainer's optimal gum barrier can be seen in FIGS. 31-32. FIG. 33 depicts the final retainer design. Once the final retainer design has been identified, an STL file containing that structure is generated and sent to a 3D printer. FIG. 34 depicts a retainer design ready to be sent to a 3D printer in STL format. FIGS. 35 and 36 illustrate the process of adding hooks for barrier connection to the designed printer. The use of the AI platform simplifies the process of adding or subtracting elements and ensures perfect matches between connection points on the retainer and barrier portions of the invention. Once the STL file has been generated, an appropriate slicer program is selected, appropriate medical-grade materials are chosen, and the STL is forwarded to the 3D printing machine. At the 3D printer, the slicer converts the STL into printable G-Code. The 3D printer then generates a custom fit retainer for the user.

(75) While the invention has been described in connection with a number of embodiments and implementations, the invention is not so limited but covers various apparent modifications and equivalent arrangements, which fall within the purview of the appended claims. Although features of the invention are expressed in certain combinations among the claims, it is contemplated that these features can be arranged in any variety and order. The invention has been described herein using specific embodiments for illustrative purposes only. It will be readily apparent to one of ordinary skill in the art, however, that the principles of the invention can be embodied in other ways. Therefore, the invention should not be limited in scope to the specific embodiments disclosed herein; it should be fully commensurate with the following claims.

Claims

1. A method of creating a retainer comprising the steps of: receiving a dental scan representing a dental structure; processing the dental scan using an artificial intelligence model, wherein the artificial intelligence model is trained on a dataset comprising a plurality of dental scans representing unique dental structures to identify one or more undercuts; and generating a digital model of a dental retainer based on the processed dental scan, wherein the digital model is optimized, through the artificial intelligence model, for the dental structure represented in the dental scan by eliminating one or more undercuts and establish borders for retainer fabrication.
 2. The method of claim 1, further comprising the steps of fabricating the dental retainer based on the digital model using a manufacturing device.
 3. The method of claim 1, wherein the artificial intelligence model comprises a machine learning model, a deep learning model, or a combination thereof.
 4. The method of claim 1, wherein the unique dental structures comprises a variety of dental structures, dental conditions, or a combination thereof.
 5. The method of claim 2, wherein the manufacturing device comprises a 3D printer.
 6. The method of claim 1, further comprising the step of adjusting the digital model based on one or more corrections received from a user or a dental professional.
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